

Unit 1: Function Analysis

Precalculus

Student

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1.1 Functions and Function Notation

Objectives

- Determine whether a relation represents a function.
- Use function notation to evaluate functions.
- Represent functions using tables, mapping diagrams, and graphs.

Relations and Functions

A **relation** is _____. The set of all first coordinates is the **domain**, and the set of all second coordinates is the **range**.

A **function** is a relation that assigns to each element in the domain _____ in the range.

Key Idea: One input $\rightarrow$ one output. Two different inputs *can* share the same output, but one input *cannot* produce two different outputs.

Example 1

Determine whether each relation represents a function.

a) $\{(1, 5), (2, 6), (3, 7), (4, 8)\}$

b) $\{(1, 5), (1, 6), (3, 7), (4, 8)\}$

Mapping Diagrams

A **mapping diagram** uses arrows to show how inputs in the domain relate to outputs in the range.

- **Function:** Each input in the domain has _____ arrow leaving it.
- **Not a Function:** If any input in the domain has _____ arrows leaving it.

Function Notation

Instead of writing y , we write _____. Read as “_____” or “the value of f at x .”

The letter f names the function. The letter x is the input.

Example 2

Evaluate $f(x) = x^2 - 3x + 5$. Find each value.

a) $f(2)$

b) $f(-1)$

c) $f(a)$

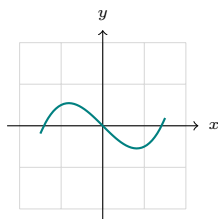
d) $f(a + h)$

Vertical Line Test (VLT)

A set of points in a coordinate plane is the graph of a function if and only if every vertical line intersects the graph in _____.

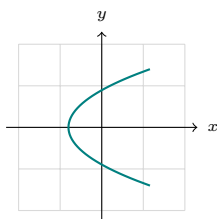
Example 3

Determine whether each graph represents a function using the vertical line test.

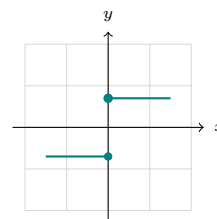


a) Wavy curve

b) Sideways parabola



c) Piecewise step

**Example 4**

Use the table of values for $f(x)$ to evaluate the expressions.

x	1	2	3	4	5
$f(x)$	8	6	7	6	8

a) Evaluate $f(1)$ b) Evaluate $f(3)$ c) Solve $f(x) = 6$ **Example 5**

The height of a baseball hit is modeled by the function $f(x) = -0.0032x^2 + x + 3$, where x is the horizontal distance from home plate in feet. Will the baseball clear a 10-foot fence located 300 feet from home plate?

One-to-One Functions

A function is **one-to-one** if each element of the range corresponds to _____ of the domain. Equivalently, if $f(a) = f(b)$, then _____. Different inputs must produce different outputs.

Example 6

Determine which of the following equations define y as a function of x .

a) $5x - 3y^2 = 9$

b) $3y^3 + x = 1$

c) $y = 5x^2 - 6x$

d) $x^2 = |y| + 6x$

Homework: Section 1.1 Homework

1.2 Domain and Range

Objectives

- Find the domain and range of functions algebraically and from graphs.
- Write intervals in interval and set-builder notation.
- Evaluate and graph piecewise-defined functions.

Domain and Range

The **domain** of a function is the set of all possible _____ (typically x) for which the function is defined and produces a real number output.

The **range** of a function is the set of all possible _____ (typically y) that the function can produce.

Representing Intervals

We can write sets of numbers using inequalities, set-builder notation, or interval notation.

- Parentheses** $()$: Used when an endpoint is _____ (strictly less than, strictly greater than, or at $\pm\infty$).
- Brackets** $[]$: Used when an endpoint is _____ (less than or equal to, or greater than or equal to).

Description	Inequality	Set-Builder	Interval
All real numbers	$-\infty < x < \infty$	$\{x \mid x \in \mathbb{R}\}$	$(-\infty, \infty)$
Greater than or equal to a	$x \geq a$	$\{x \mid x \geq a\}$	$[a, \infty)$
Strictly between a and b	$a < x < b$	$\{x \mid a < x < b\}$	(a, b)
Closed interval between a and b	$a \leq x \leq b$	$\{x \mid a \leq x \leq b\}$	$[a, b]$

Algebraic Domain Restrictions

Unless specified otherwise, we assume the domain is all real numbers. We must restrict the domain algebraically in two common cases:

1. **Division by Zero:** The denominator of a fraction cannot equal zero. If $f(x) = \frac{g(x)}{h(x)}$, we set _____.
2. **Even Roots of Negatives:** The radicand (expression inside) of an even-index root must be non-negative. If $f(x) = \sqrt[k]{g(x)}$, we set _____.

Note: Polynomials and odd roots have no algebraic restrictions, so their domain is $(-\infty, \infty)$.

Example 1

Find the domain of each function in interval notation.

a) $f(x) = \frac{x+1}{2-x}$

b) $g(x) = \sqrt{7-x}$

c) $h(x) = \frac{5}{\sqrt{x-3}}$

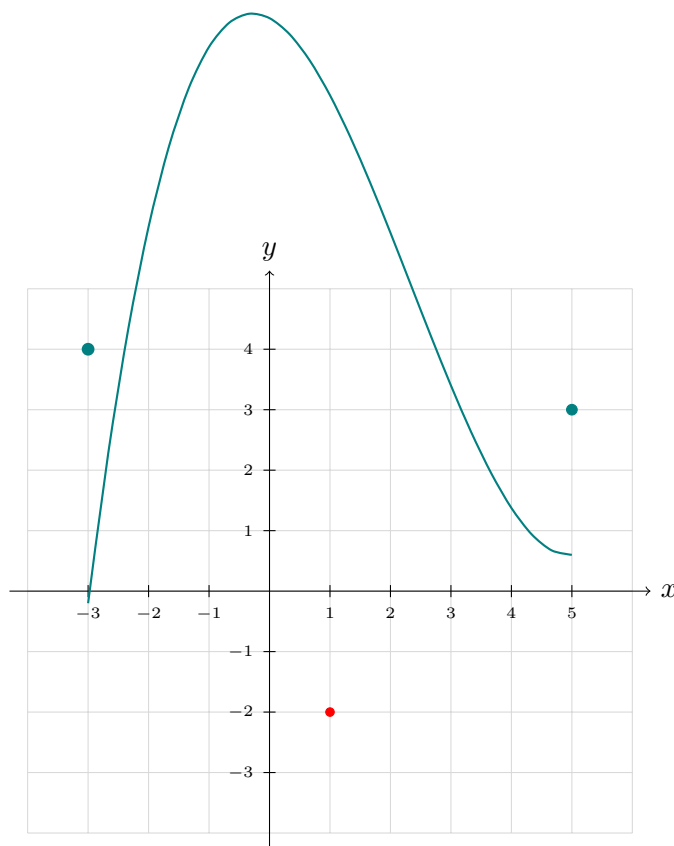
Domain and Range from Graphs

To determine domain and range from a graph:

- **Domain:** Scan the graph horizontally (left to right) along the x -axis.
- **Range:** Scan the graph vertically (bottom to top) along the y -axis. Check local peaks and valleys!
- **Endpoints:** A **closed circle** • indicates the value is _____ ($[]$). An **open circle** ○ indicates the value is _____ ($()$).

Example 2

State the domain and range of the function shown in the graph below in interval notation.

**Piecewise Functions**

A **piecewise function** is defined by more than one formula, where each formula applies to a different part of the domain.

To evaluate $f(a)$:

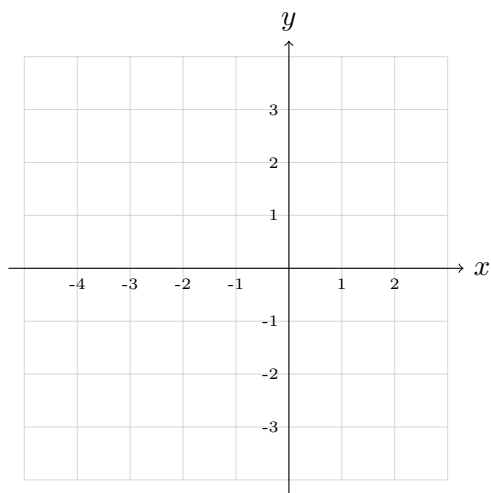
1. Check which interval a falls into.
2. Plug a into **only** the equation for that interval.

Example 3

Consider the piecewise function:

$$f(x) = \begin{cases} x + 1, & \text{if } x < -2 \\ -2x - 3, & \text{if } x \geq -2 \end{cases}$$

- a) Evaluate $f(-3)$. b) Evaluate $f(-2)$. c) Graph the function and state its range.

**Example 4**

A cellular provider charges \$30 per month for a basic plan, plus \$10 per gigabyte (GB) of data used, capped at a maximum of 5 GB.

- a) Write the monthly cost $C(g)$ as a function of data usage g .
 b) State the domain and range of this function in this real-world context.

Homework: Section 1.2 Homework

1.3 Rates of Change and Behavior of Graphs

Objectives

- Calculate the average rate of change of a function.
- Identify intervals where a function is increasing, decreasing, or constant.
- Locate local and absolute extrema on a graph.

Average Rate of Change

The **average rate of change** between two input values is the change in the output values divided by the change in the input values.

$$\text{Average Rate of Change} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

- **Geometric Interpretation:** The average rate of change represents the _____ passing through the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ on the graph of the function.
- **Units:** The units of the average rate of change are always _____ (e.g., miles per hour, dollars per year).

Example 1

The table shows the average price of a gallon of gasoline in the US between 2005 and 2020. Find the average rate of change of gas prices between the given years.

Year (t)	2005	2008	2012	2016	2020
Price ($P(t)$)	\$2.30	\$3.27	\$3.64	\$2.14	\$2.60

a) Between 2005 and 2012

b) Between 2012 and 2020

Example 2

Find the average rate of change of $f(x) = x^2 - \frac{1}{x}$ on the interval $[2, 4]$.

Increasing, Decreasing, and Constant Behavior

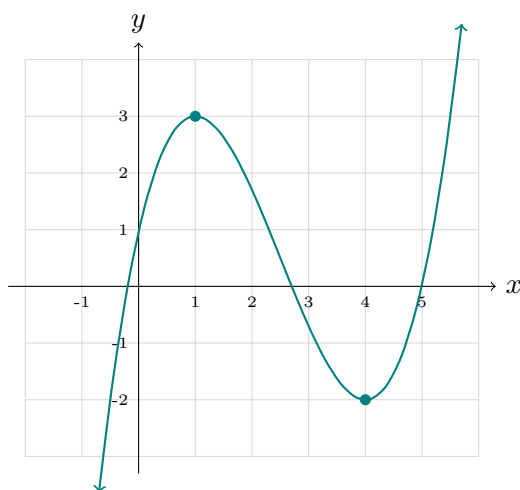
Read the graph from **left to right**:

- **Increasing:** Graph rises from left to right. Outputs increase as inputs increase: $f(x_1) < f(x_2)$ when $x_1 < x_2$.
- **Decreasing:** Graph falls from left to right. Outputs decrease as inputs increase: $f(x_1) > f(x_2)$ when $x_1 < x_2$.
- **Constant:** Graph is flat. Outputs do not change: $f(x_1) = f(x_2)$ for all x .

Critical Rule: When reporting intervals where a function is increasing, decreasing, or constant, always write the intervals in terms of _____ using open intervals (parentheses).

Example 3

Identify the intervals where the function graphed below is increasing and/or decreasing.

**Local and Absolute Extrema**

- A **local (relative) maximum** is a point higher than any nearby points (a hill peak).
- A **local (relative) minimum** is a point lower than any nearby points (a valley bottom).
- An **absolute (global) maximum** is the highest output value over the function's entire domain.
- An **absolute (global) minimum** is the lowest output value over the function's entire domain.

Domain Rule: On a closed interval $[a, b]$, absolute extrema must exist and can occur at _____. However, local extrema can only occur on the interior of the domain, never at endpoints.

Example 4

Using the graph from Example 3, identify all local and absolute extrema on the domain $(-\infty, \infty)$.

Example 5

Suppose the domain of the function from Example 3 is restricted to the closed interval $[0, 5]$, where $f(0) = 1$ and $f(5) = 1$ (using the endpoints). Find the local and absolute extrema.

Homework: Section 1.3 Homework

1.4 Composition of Functions**Objectives**

- Combine functions algebraically and determine their domains.
- Evaluate composite functions numerically and algebraically.
- Find the domain of a composite function.
- Decompose a composite function into component functions.

Arithmetic Operations on Functions

We can combine functions using standard arithmetic operations. Let f and g be functions:

- **Sum:** $(f + g)(x) = f(x) + g(x)$
- **Difference:** $(f - g)(x) = f(x) - g(x)$
- **Product:** $(f \cdot g)(x) = f(x) \cdot g(x)$
- **Quotient:** $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$

Overlapping Domains: The domain of $f \pm g$, $f \cdot g$, and $\frac{f}{g}$ is the _____ (overlap) of the domain of f and the domain of g . For the quotient function, we must also exclude any values of x where _____.

Example 1

Given $f(x) = 3x^2 + 7x$ and $g(x) = 2x^2 - x - 1$, find $(f - g)(x)$.

Example 2

Given $f(x) = 3x^2 - 2x + 1$ and $g(x) = x - 4$, find $\left(\frac{f}{g}\right)(x)$ and state its domain in interval notation.

Composition of Functions

When we compose functions, we nest one function inside another:

- **f composed with g :** $(f \circ g)(x) = f(g(x))$ (read as “ f of g of x ”)
- **g composed with f :** $(g \circ f)(x) = g(f(x))$ (read as “ g of f of x ”)

Order Matters: Composition is not commutative: in general, $(f \circ g)(x) \neq (g \circ f)(x)$.

Strategy: Always evaluate a composition from the _____.

Example 3

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, evaluate the following expressions:

a) $f(g(1))$

b) $g(f(2))$

Example 4

Given $f(x) = x - 3$ and $g(x) = 2x^2 - 1$, find the following algebraic composite expressions:

a) $(f \circ g)(x)$

b) $(g \circ f)(x)$

Domain of a Composite Function

To find the domain of the composite function $f(g(x))$:

1. Find the domain of the **inner function** $g(x)$. Exclude any restricted values.
2. Form the **composite expression** $f(g(x))$, simplify, and identify any additional restrictions on this final expression.
3. Combine the exclusions. The final domain is where both conditions are satisfied.

Example 5

Find the domain of $(f \circ g)(x)$ given $f(x) = \frac{1}{x+2}$ and $g(x) = \frac{4}{x-1}$.

Decomposition of Functions

To **decompose** a complex function $h(x)$ means to find an outer function $f(x)$ and an inner function $g(x)$ such that $h(x) = f(g(x))$.

Look for group expressions inside parentheses, square roots, denominators, or exponents.

Example 6

Find functions $f(x)$ and $g(x)$ such that $h(x) = f(g(x))$ for each function.

a) $h(x) = \sqrt{5x-4}$

b) $h(x) = \frac{4}{(x+2)^3}$

Homework: Section 1.4 Homework

1.5 Transformation of Functions**Objectives**

- Identify translations, reflections, and dilations from their equations.
- Apply the correct order of transformations to graph functions.
- Classify functions as even, odd, or neither algebraically and graphically.

Parent Functions

You must be familiar with the following basic parent functions:

- | | | |
|----------------------------------|---------------------------------------|--|
| • Linear: $f(x) = x$ | • Cubic: $f(x) = x^3$ | • Square Root: $f(x) = \sqrt{x}$ |
| • Quadratic: $f(x) = x^2$ | • Absolute Value: $f(x) = x $ | • Cube Root: $f(x) = \sqrt[3]{x}$ |

Three Flavors of Transformation

Let $f(x)$ be a parent function. We can transform it in three ways:

1. **Translation (Shift):** Slides the graph horizontally by h or vertically by k .

$$g(x) = f(x - h) + k$$

Note: $f(x - h)$ shifts _____ (opposite of sign) and $f(x) + k$ shifts _____

(behaves as expected).

2. **Reflection:** Flips the graph across the axes.

- $-f(x)$: Negates y -values $\rightarrow$ reflects over the _____.
- $f(-x)$: Negates x -values $\rightarrow$ reflects over the _____.

3. **Dilation (Stretch/Compression):** Changes the size/shape of the graph.

- $c \cdot f(x)$ (vertical): Stretch if $c > 1$, compress if $0 < c < 1$.
- $f(kx)$ (horizontal): Compress if $k > 1$, stretch if $0 < k < 1$ by a factor of $\frac{1}{k}$.

Core Principle: Changes ****outside**** affect y (vertical, normal behavior). Changes ****inside**** affect x (horizontal, opposite behavior).

Example 1

Describe the transformations from the parent function for each of the following:

a) $g(x) = |x - 1| + 12$

b) $h(x) = -\sqrt{x + 8}$

c) $f(x) = 3(x - 7)^3$

d) $h(x) = \sqrt[3]{2x - 9}$

Order of Transformations: D.R.S.

When multiple transformations are applied, follow the **D.R.S.** order:

$$\mathbf{D} \rightarrow \mathbf{R} \rightarrow \mathbf{S} \quad (\text{Dilate} \rightarrow \text{Reflect} \rightarrow \text{Shift})$$

Applying transformations in this order ensures that shifts are performed on the dilated and reflected graph correctly.

Even and Odd Functions

- **Even Function:** Symmetric with respect to the _____. Algebraic test: _____.
- **Odd Function:** Symmetric with respect to the _____. Algebraic test: _____.

Example 2

Classify each function as even, odd, or neither algebraically.

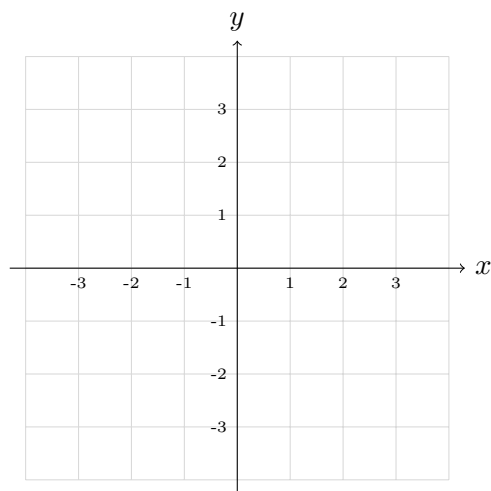
a) $f(x) = x^3 - x$

b) $g(x) = x^2 + 2$

c) $h(x) = x^3 - 1$

Example 3

Describe the transformations, map the points, and graph $y = -2|x| + 1$.



Transformations (D.R.S.):

- D: _____
- R: _____
- S: _____

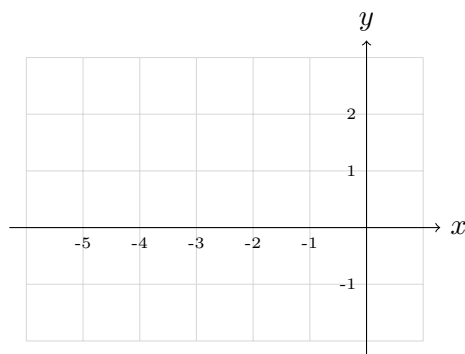
Mapping

Table:

x	Parent $ x $	Transformed
-2	2	_____
-1	1	_____
0	0	_____
1	1	_____
2	2	_____

Example 4

Describe the transformations, map the points, and graph $y = \sqrt{-2(x+3)} - 1$.



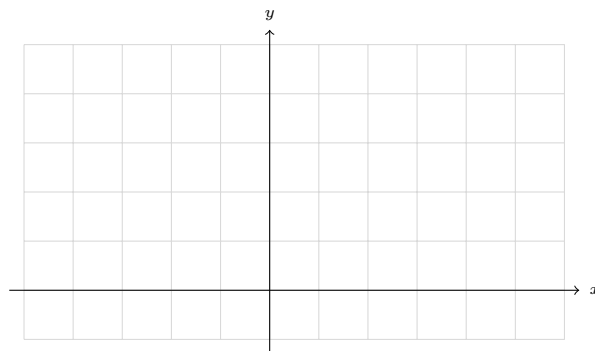
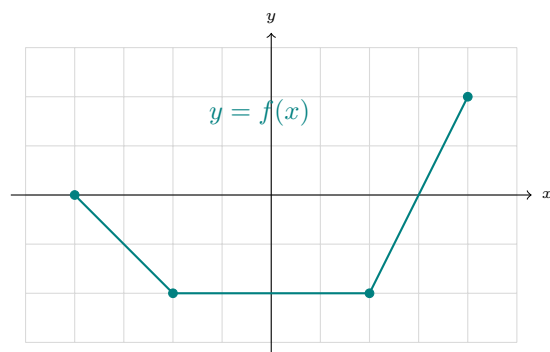
Transformations (D.R.S.):

- D: _____
- R: _____
- S: _____

Mapping Table:	Parent $(x, \sqrt{x})$	Transformed
	(0, 0)	_____
	(1, 1)	_____
	(4, 2)	_____

Example 5

Given the graph of $y = f(x)$ below, graph $g(x) = -\frac{1}{2}f(x-1) + 3$.



Homework: Section 1.5 Homework

1.6 Absolute Value Functions**Objectives**

- Graph absolute value functions using vertex and transformations.
- Solve absolute value equations algebraically.

- Solve absolute value inequalities and graph their solutions.

The Absolute Value Function

The parent function $f(x) = |x|$ represents the distance of x from the origin. It is defined piecewise as:

$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

- **Domain:** $(-\infty, \infty)$
 - **Vertex:** The corner point where the graph changes direction, which is at $(0, 0)$.
- Range:** $[0, \infty)$

Graphing with Transformations

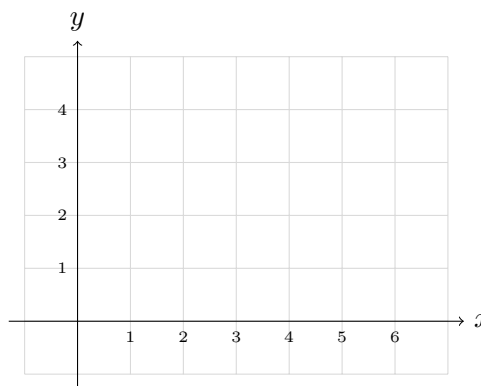
The general form of a transformed absolute value function is:

$$f(x) = a|x - h| + k$$

- **Vertex:** Shifted to _____.
- **Orientation:** Opens up if $a > 0$; opens down if $a < 0$.
- **Branch Slopes:** The right branch has slope a , and the left branch has slope $-a$.

Example 1

Describe the transformations and graph the function $f(x) = -2|x - 3| + 4$.



Transformations:

- Vertex: _____
- Shape: opens _____
- Branch Slopes: _____ (right), _____ (left)

Solving Absolute Value Equations

To solve an equation containing an absolute value, first **isolate** the absolute value expression:

$$|u| = c$$

- If $c < 0$: There is _____ (distance cannot be negative).
- If $c = 0$: Solve the single case: _____.
- If $c > 0$: Split into two equations: _____.

Example 2

Solve the equation: $2|3x - 1| + 5 = 15$

Solving Absolute Value Inequalities

For any algebraic expression u and a positive real number c :

- **Less-Than** (“Less-and”):

$$|u| < c \implies \underline{\hspace{2cm}} \underline{\hspace{2cm}}$$

This yields a single, bounded interval.

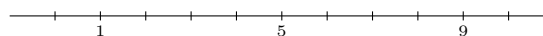
- **Greater-Than** (“Great-or”):

$$|u| > c \implies \underline{\hspace{2cm}} \underline{\hspace{2cm}}$$

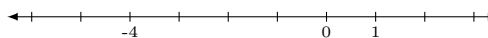
This yields a union of two unbounded intervals.

Example 3

Solve the inequality and graph the solution: $|x - 5| < 4$

**Example 4**

Solve the inequality and graph the solution: $|2x + 3| \geq 5$



Homework: Section 1.6 Homework

1.7 Inverse Functions**Objectives**

- Verify inverse functions using composition.
- Find inverse functions algebraically.
- Identify one-to-one functions using the horizontal line test.
- Restrict domains to find inverse functions.

Inverse Functions Swap Domain & Range

An **inverse function** undoes the action of the original function. If f maps a to b , then the inverse function f^{-1} maps b to a .

- **Notation:** The inverse of $f(x)$ is written as $f^{-1}(x)$ (read as “ f inverse of x ”). Note: this is **not** an exponent, i.e., $f^{-1}(x) \neq \frac{1}{f(x)}$.

- **Domain and Range:**

$$\text{Domain of } f^{-1} = \text{Range of } f \quad \text{and} \quad \text{Range of } f^{-1} = \text{Domain of } f$$

- **Coordinates:** If the point (a, b) lies on the graph of $f(x)$, then the point _____ lies on the graph of $f^{-1}(x)$.

How to Find an Inverse Function Algebraically

To find the formula for the inverse function $f^{-1}(x)$ of a one-to-one function $f(x)$:

1. Replace $f(x)$ with y .
2. Interchange the variables x and y .
3. Solve the resulting equation for y .
4. Replace y with $f^{-1}(x)$ (and state any domain restrictions).

Example 1

Find the inverse function of $f(x) = x^3 - 4$.

Example 2

Find the inverse function of $f(x) = \frac{3x + 1}{x - 2}$.

Example 3

Verify that $f(x) = \frac{4}{x-3}$ and $g(x) = \frac{4}{x} + 3$ are inverse functions of each other by showing that $f(g(x)) = x$ and $g(f(x)) = x$.

Graphical Relationship of Inverses

The graphs of $f(x)$ and $f^{-1}(x)$ are reflections of each other across the identity line _____. Folding the coordinate plane along this line places the graph of the function exactly onto its inverse.

Horizontal Line Test & One-to-One

- A function is **one-to-one** if each output in the range corresponds to _____ in the domain.
- **Horizontal Line Test (HLT)**: A function has a function inverse if and only if its graph intersects every horizontal line in _____.

Example 4

Determine whether each function is one-to-one and thus has an inverse function.

a) $f(x) = x^2 - 3$

b) $g(x) = x^3$

c) $h(x) = \sqrt{x}$

Restricting Domains to Force Invertibility

If a function is not one-to-one, it does not have an inverse function. However, we can **restrict the domain** of the original function to an interval where it is strictly increasing or decreasing (and thus one-to-one) to define an inverse function over that interval.

Example 5

The function $f(x) = (x-2)^2$ is not one-to-one. Restrict its domain to $[2, \infty)$ and find its inverse function.